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# End - End Communication at Transport Layer.

(a) Implement echo client server using TCP / UDP sockets

(b) Implement a chat program using socket programming

TCP:

```
import socket
import threading
# server
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM
    server.bind(('127.0.0.1', 500))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client."
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"server received: {data}".encode()
    conn.close()
    server.close()
```

```
# client
def start_client():
    client = socket.socket(socket.AF_INET, socket.
                           SOCK_STREAM)

    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit):")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

# main
if __name__ == "__main__":
    # Run server ing bg thread
    server_thread = threading.Thread(target=start_server,
                                     daemon=True)

    server_thread.start()
    # Run client in main thread
    start_client()
```

O/P:

```
[SERVER] Listening on 127.0.0.1:5000
[CLIENT] Connected to server
[SERVER] Connected by ('127.0.0.1', 60530)
Enter message (or 'quit' to exit): Hello
[SERVER] Received: Hello
[SERVER RESPONSE] Server received: Hello
Enter message (or 'quit' to exit): Everyone
[SERVER] Received: Everyone
[SERVER RESPONSE] Server received: Everyone
Enter message (or 'quit' to exit): quit
[SERVER] Connection closed by client.
```

UDP:

```
import socket
import threading

# server
def udp_server():
    s=socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended
                the connection.")

            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr) # Echo back

    s.close()
    print("[SERVER] stopped.")

# client
def udp_client():
    c=socket.socket(socket.AF_INET, socket.
                    SOCK_DGRAM)
    server = ('127.0.0.1', 12000)

    while True:
        msg = input("Enter message (or 'quit' to
                    stop): ")

        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connection...") break
```

```
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[CLIENT] Stopped.")

# main
if __name__ == "__main__":
    threading.Thread(target=udp_server,
                     daemon=True).start()
    udp_client()
```

o/p:

```
[SERVER] Running on 127.0.0.1:12000
Enter message (or 'quit' to stop): HI
[SERVER] Received from ('127.0.0.1', 53267): HI
Echo from Server: HI
Enter message (or 'quit' to stop): Everyone
[SERVER] Received from ('127.0.0.1', 53267): Everyone
Echo from Server: Everyone
Enter message (or 'quit' to stop): quit
[CLIENT] Closing connection...
[CLIENT] Stopped.
```

RESULT:

Thus End-End Communication was executed.